

EDA Notebook

0. Setup Environment

```
# DO NOT MODIFY THE CODE IN THIS CELL
!pip install -q utstd

from utstd.folders import *
from utstd.ipyrenders import *

at = AtFolder(
    course_code=36106,
    assignment="AT3",
)
at.run()

import warnings
warnings.simplefilter(action='ignore')
```

```
12.9/12.9 MB 38.6 MB/s eta
0:00:00
```

```
4.9/4.9 MB 72.5 MB/s eta
0:00:00
```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

umap-learn 0.5.12 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2 which is incompatible.

hdbscan 0.8.42 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2 which is incompatible.

Mounted at /content/gdrive

You can now save your data files in:
/content/gdrive/MyDrive/36106/assignment/AT3/data

Student Information

```
# <Student to fill this section>
group_name = "Group 20"
student_name = "Ratnadeep Patra"
student_id = "26294153"

# Do not modify this code
print_tile(size="h1", key='group_name', value=group_name)
```

```
<IPython.core.display.HTML object>

# Do not modify this code
print_tile(size="h1", key='student_name', value=student_name)

<IPython.core.display.HTML object>

# Do not modify this code
print_tile(size="h1", key='student_id', value=student_id)

<IPython.core.display.HTML object>
```

0. Python Packages

0.a Install Additional Packages

If you are using additional packages, you need to install them here using the command: `! pip install <package_name>`

```
# <Student to fill this section>
```

0.b Import Packages

```
# <Student to fill this section>
import pandas as pd
import altair as alt

import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import os
```

B. Data Understanding

```
# Do not modify this code
try:
    df = pd.read_csv(at.folder_path / "product_category.csv")
except Exception as e:
    print(e)
```

B.1 Explore Dataset

```
pd.set_option('display.max_columns', None) # Forcing pandas to display
all columns
pd.set_option('display.max_rows', None) # Forcing pandas to display
all rows
```

```
df.describe(include='all').T
```

```
{
  "summary": {
    "name": "df",
    "rows": 2,
    "fields": [
      {
        "column": "count",
        "properties": {
          "dtype": "date",
          "min": "4",
          "max": "4",
          "num_unique_values": 1,
          "samples": [
            "4"
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "unique",
        "properties": {
          "dtype": "date",
          "min": 4,
          "max": 4,
          "num_unique_values": 1,
          "samples": [
            4
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "top",
        "properties": {
          "dtype": "string",
          "num_unique_values": 2,
          "samples": [
            "Bikes"
          ],
          "semantic_type": "",
          "description": ""
        }
      },
      {
        "column": "freq",
        "properties": {
          "dtype": "date",
          "min": "1",
          "max": "1",
          "num_unique_values": 1,
          "samples": [
            "1"
          ],
          "semantic_type": "",
          "description": ""
        }
      }
    ],
    "type": "dataframe"
  }
}
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 4 entries, 0 to 3
```

Data columns (total 2 columns):

#	Column	Non-Null Count	Dtype
0	product_category_id	4 non-null	object
1	name	4 non-null	object

```
dtypes: object(2)
```

```
memory usage: 196.0+ bytes
```

```
df[ 'name' ]
```

```
0 Bikes
```

1 Components

2 Clothing

3 Accessories

```
Name: name, dtype: object
```

<Student to fill this section>

```
dataset insights = ""
```

This dataset contains 4 entries and 2 columns - 'product_category_id' and 'name'. Both columns are of object (string) data type. Each row represents a

unique product category. The four categories are: Bikes, Components, Clothing, and Accessories.

|| || ||

```
# Do not modify this code
print_tile(size="h3", key='dataset_insights', value=dataset_insights)

<IPython.core.display.HTML object>
```

B.2 Explore Feature of Interest product_category_id

```
for csv_file in os.listdir(at.folder_path):
    if csv_file.endswith('.csv'):
        try:
            if 'product_category_id' in pd.read_csv(at.folder_path /
csv_file).columns:
                print(csv_file)
            except UnicodeDecodeError:
                print(f"Warning: Could not read {csv_file} with default
encoding. Skipping or consider specifying encoding.")
            except Exception as e:
                print(f"Error reading {csv_file}: {e}")

product_category.csv
product_sub_category.csv

# <Student to fill this section>
feature_1_insights = """
The 'product_category_id' column serves as a unique identifier for
each product category within this dataset. Its primary utility lies in
its ability to
act as a foreign key for joining with 'product_sub_category.csv'. This
allows for more comprehensive investigations into the hierarchical
structure of
products, enabling analysis at a more granular level. By joining these
datasets, one can explore relationships between broader categories and
their
specific sub-categories, or analyze sales and product performance
across different categorical levels.
"""

# Do not modify this code
print_tile(size="h3", key='feature_1_insights',
value=feature_1_insights)

<IPython.core.display.HTML object>
```